

Part C) Complete the following table and select which software process model is appropriate for the given concern. You must provide a reason for your selection.

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Concern	Choose one
C1) The target system is intended for safety-critical operations. Reason: <i>waterfall more safety because the requirements must be complete to go to next step of project.</i>	☒ Waterfall ☐ Agile ☐ Component-based
C2) The clients need to see prototypes of the developed system and provide early feedback to the software engineers. Reason: <i>the Agile make the customer one of the team to take feedback in every (version).</i>	☐ Waterfall ☒ Agile ☐ Component-based

Part D) Answer the following questions that are related to Software Engineering Ethics and Professional Practices (SEEPP):

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D1) [2 marks] One principle that SEEPP covers is Public, which states that software engineers shall act consistently in the public interest. Give two examples of adhering to this principle.

- * as software engineering we must develop and create APP that help people
- * we must not use the harm or ifedee on people mainly or use them to do something

D2) [2 marks] In the following table, indicate which principle is best submitted to the provided example. Principles that can be considered include:

1. Management: promote an ethical approach to managing software development.
2. Profession: advance the profession's integrity consistent with the public interest.
3. Judgment: maintain integrity, morals, and independence in their professional judgment.
4. Self: participate in lifelong learning of the practice of their profession.

Example	Principle
D2.1) A software engineer must <u>ensure</u> a <u>realistic</u> estimation of cost and plan on all projects.	<u>management</u>
D2.2) A software engineer shouldn't <u>illegally</u> <u>obtain</u> a software application for <u>any</u> purpose.	<u>Profession</u>

Part A) In the following multiple-choice items, put a circle around the letter of the correct choice. There is only one correct answer per item.

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A1) The following is an example of a generic type of software product:

a. patient registration application of a private hospital

b. Microsoft PowerPoint

c. a stock prediction application for investment in a company

A2) Motivation behind software engineering includes:

a. to rapidly produce large software system

b. to handle development and maintenance cost

c. to distinguish market needs of software

A3) The following is NOT part of the software engineering process activity:

a. software design

b. software specification

c. software management

A4) In the Scrum management framework, the inputs of sprint planning include:

a. team capacity

b. sprint goal

c. sprint backlog

A5) Product quality relies on process quality, management quality, and:

a. development technology

b. fulfilment of clients' requirements

c. short delivery window

A6) One of the essential software design activities is:

a. requirements design

b. architectural design

c. test design

Part B) Determine whether the following statements are valid or not. Select true or false and then support your answer with a reason regardless of the state.

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B1) It is considered that software development has more cost (timewise) than software maintenance.

Reason:

True | False

the maintenance took more time than development and testing

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B2) The initial estimation of engineers' efforts toward software development is always accurate.

Reason:

True | False